

Development of Head and Neck

Skull development:

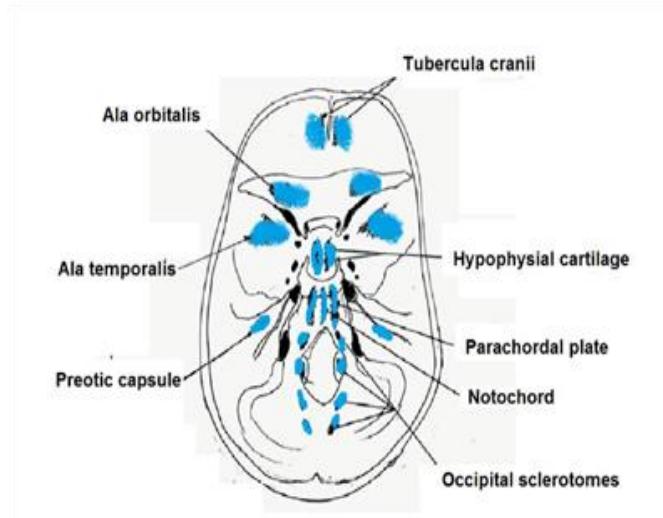
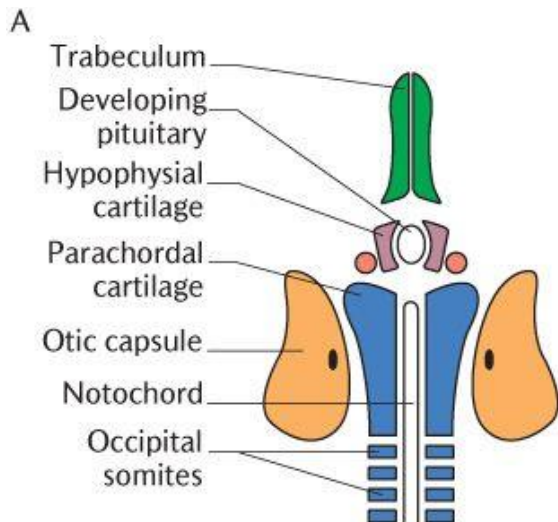
- The skull development is divided into two parts:
 - The neurocranium consists of bones of skull around the brain.
 - The viscerocranium, which consists of the bones of the face.
- The **neurocranium** is divided into two portions:
 - The **membranous part**, or **membranocranium**
 - The **cartilaginous part**, or **chondrocranium**.

Membranocranium (Membranous neurocranium):

- The is a part of the skull produced by intramembranous ossification.
- It is derived from neural crest cells and paraxial mesoderm.
- Mesenchyme from these two sources invests the brain and undergoes intramembranous ossification.
- The result is formation of several flat bones of skull vault connected by skull sutures.

Chondrocranium (cartilaginous neurocranium):

- Bones of the base of the skull (floor of the neurocranium) are produced by endochondral bone formation (cartilaginous ossification).
 - The following cartilages will give rise to bones of skull base:
 - **Two tubercula cranii** give the ethmoidal bone.
 - **Two hypophysial cartilages** form the body of sphenoid.
 - **Two ala orbitalis** form the lesser wing of sphenoid.
 - **Two ala temporalis** form the greater wing of sphenoid.
 - **Two preotic capsules** form the petrous temporal bone.
 - **Parachordal plates** and **four pairs of occipital sclerotomes** form the basilar part of the occipital bone.
- **Congenital anomalies of skull:** **Craniosynostosis** premature cranial suture fusion.
Cranioschisis: a large defect in the skull. **Cranial meningocele, encephalocele:** herniation of meninges or brain tissue through a defect in the skull

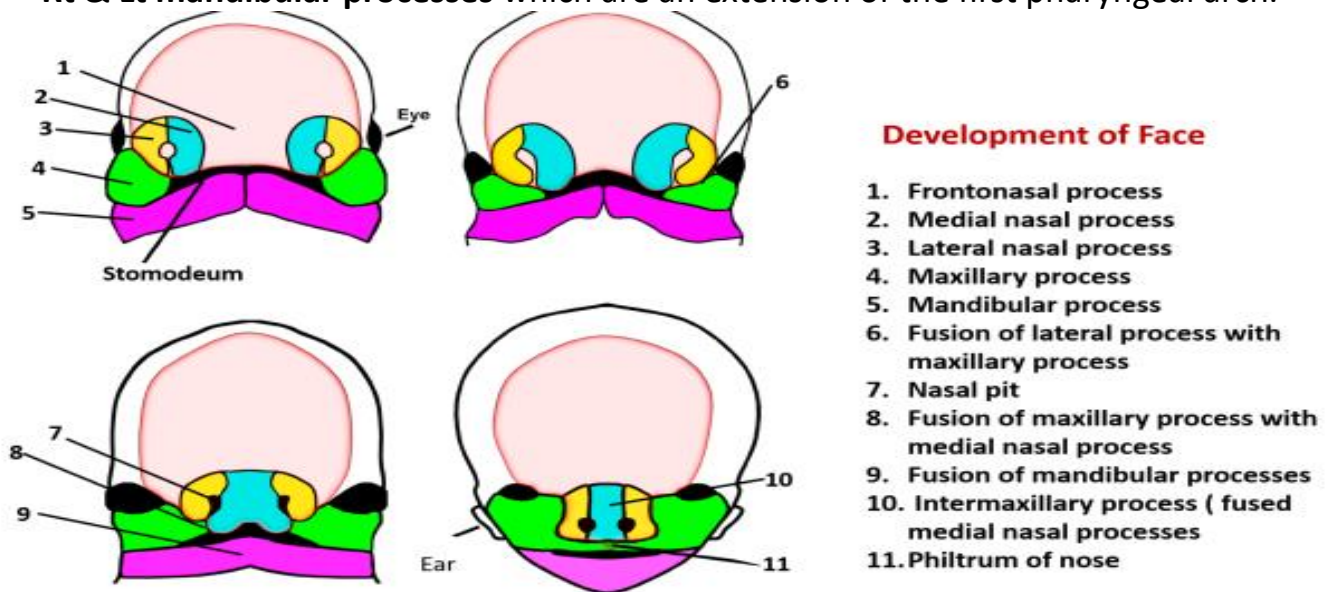


❑ **Viscerocranium:** forms the skeleton of the face. The face develops from 3 sources that surround the stomodeum (primitive mouth):

Frontonasal process is formed by proliferation of mesenchyme ventral to forebrain.

Rt & Lt maxillary processes which are an extension of the first pharyngeal arch.

Rt & Lt mandibular processes which are an extension of the first pharyngeal arch.



Facial muscles are derived from the 2nd pharyngeal arch.

Derivatives:

- **Intermaxillary segment** gives rise to: Philtrum, Median part of upper jaw carrying 4 incisors, Primitive palate & Primitive nasal septum.
- **Maxillary process** gives rise to: Upper lip except philtrum, Upper jaw except the part carrying incisors, Definitive palate (palatine bone), Maxilla, zygomatic bone, lacrimal bone, squamous temporal bones, vomer and palatine bone.
- **Mandibular process** gives rise to: Whole lower lip, Whole lower jaw, Part of cheek
- **Nasolacrimal cord** is canalized to form: Nasolacrimal duct & Lacrimal sac.

Development of the palate:

- The primary palate is derived from the intermaxillary segment.
- The definitive palate is formed by inward growth of two horizontal palatine shelves from the maxillary prominences.
- These palatine shelves fuse together forming the secondary palate.
- **Y-shaped** fusion between the two palatine shelves, triangular primary palate and the nasal septum.

Development of the nose:

- Frontonasal prominence gives the bridge of nose and nasal septum.
- Medial nasal fold gives crest and tip of nose.
- Lateral nasal fold gives alae of the nose.

Development of the eye:

- Optic vesicle (optic cup) which is derived from neuroectodermal cells of forebrain gives rise to retina and iris. Optic stalk is transformed into optic nerve.
- The lens placode (lens vesicle) which is formed of cells of surface ectoderm, in contact with the optic vesicle.
- Sclera and choroid are mesenchymal in origin.
- Cornea is ectodermal in origin.

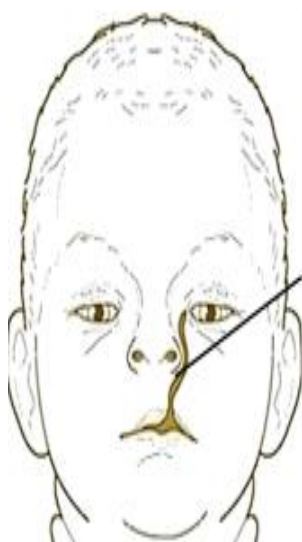
Development of the ear:

- I. **External ear:** The external auditory meatus develops from the **first pharyngeal cleft**. The auricle develops from six mesenchymal thickenings/proliferations of the 1st and 2nd pharyngeal arches, surrounding the 1st pharyngeal cleft.
- II. **Middle ear: tympanic cavity & auditory tube** are derived from the endoderm of the **first pharyngeal pouch**. **3 Ossicles** are mesodermal in origin; **malleus** and **incus** are derived from the **1st pharyngeal arch**, and **stapes** is derived from the **2nd arch**.
- III. **Internal ear:** it develops from the otic placode (ectodermal thickening) on each side of the rhombencephalon. These otic placodes invaginate rapidly to form the otic vesicles.

Congenital Anomalies:

- **Oblique facial cleft:** due to failure of fusion between maxillary process & frontonasal process.
- **Atresia** of nasolacrimal duct due to failure of canalization of nasolacrimal cord.
- **Cleft lip (hare lip):**
 - Unilateral type: due to failed fusion between maxillary process & medial nasal fold at one side.

- Bilateral type: the same cause on both sides, the philtrum is free.
- Median type: failure of development of the medial nasal folds.
- **Cleft palate:**
 - Anterior cleft palate: due to failure of fusion between secondary palate & primitive palate.
 - Posterior cleft palate: due to failed fusion between the two horizontal palatine processes with nasal septum.
 - Complete anterior and posterior cleft palate.



Oblique facial cleft



Unilateral cleft lip



Bilateral cleft lip



Median cleft lip



Normal Palate



Unilateral Cleft Lip and Palate



Bilateral Cleft Lip and Palate

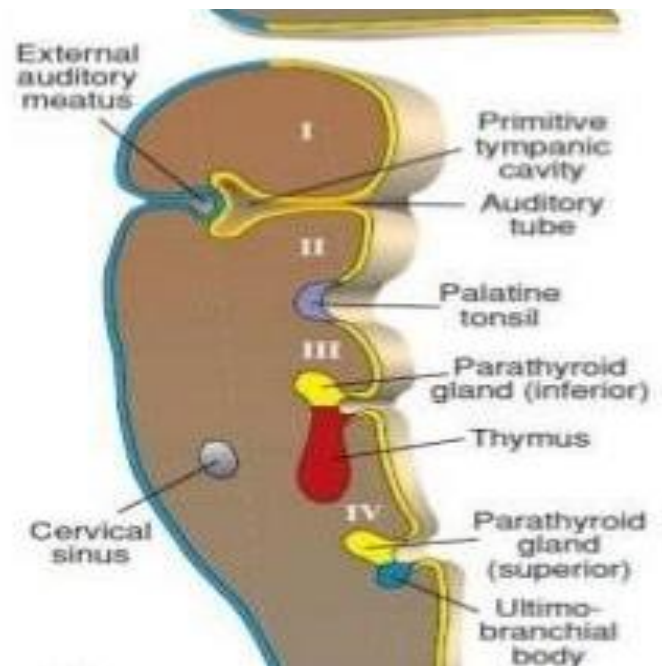
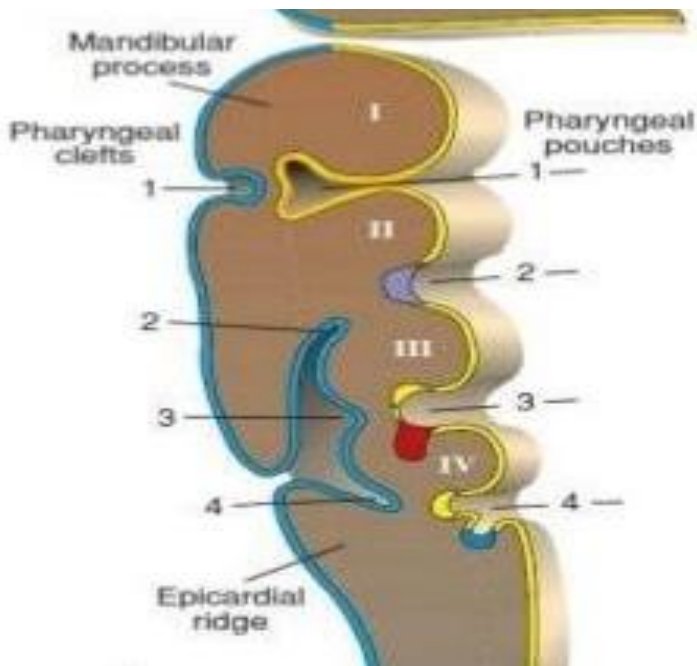
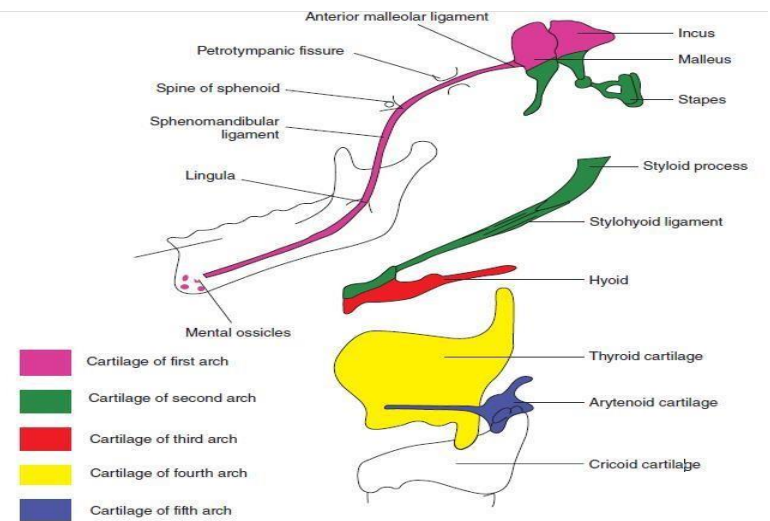
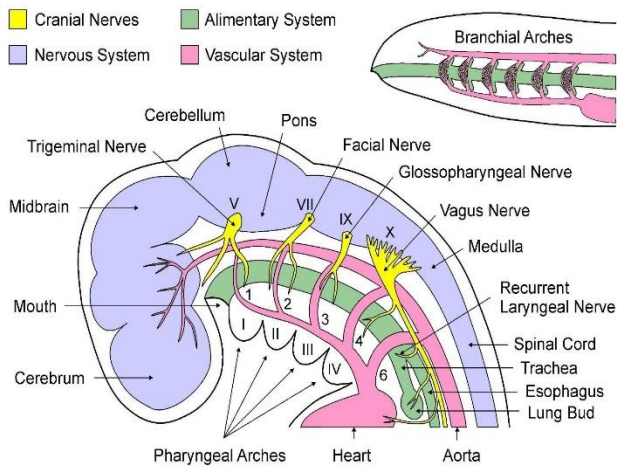
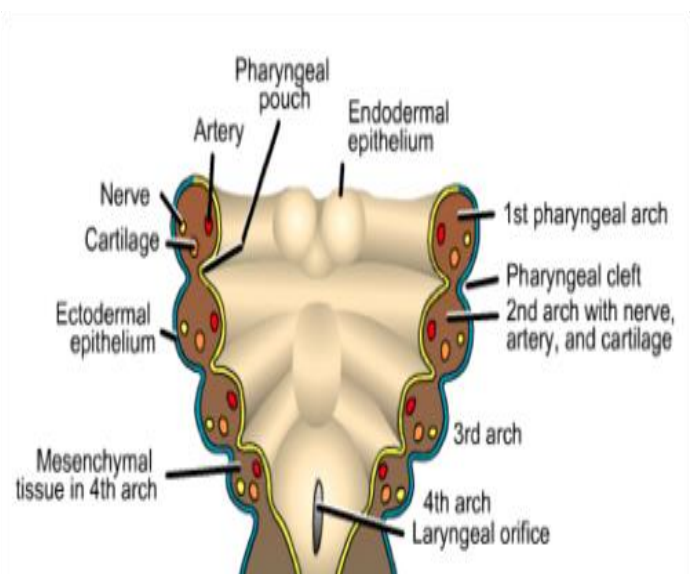
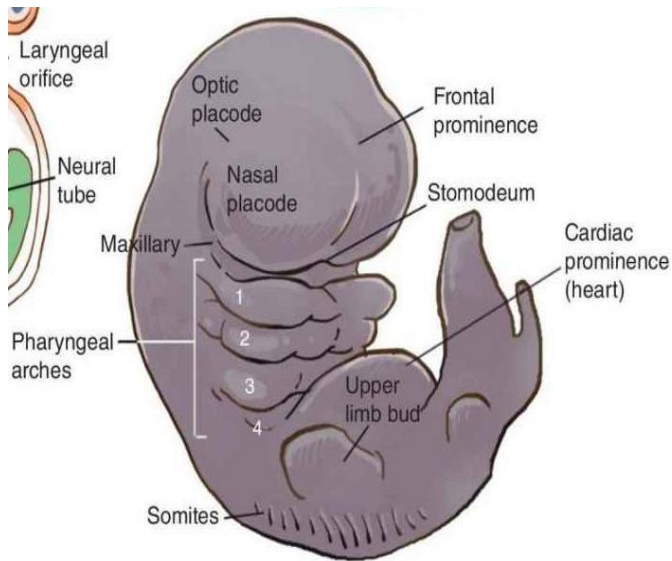


Branchial apparatus

- It consists of: Pharyngeal (Branchial) arches, Pharyngeal pouches, Pharyngeal clefts & Pharyngeal Membranes.
- **Pharyngeal (Branchial) arches:** They are six mesodermal thickenings that appear in the side wall of primitive pharynx caudal to stomodeum. The 5th arch is rudimentary and rapidly disappears.
- Each pharyngeal arch is associated with specific structures within the head and neck.
- **Structure of each pharyngeal arch:** Skeletal element, Muscle element, Cranial nerve element & Vascular element.

Arch	Skeletal derivatives	Muscular derivatives	Vascular supply	NERVE SUPPLY
1 st	(Mandibular cartilage of Meckel) - Mandible - Future maxilla, zygomatic bone, part of temporal bone. - Malleus and incus - Sphenomandibular ligament - Anterior ligament of malleus -	- Muscles of mastication: temporalis, masseter, lateral and medial pterygoids - Anterior belly of digastric - Mylohyoid, tensor palati, tensor tympani.	maxillary artery,	Mandibular nerve (V)
2 nd	(Hyoid cartilage of Reichert) - Stapes. - Styloid process. - Lesser cornu and Upper part of body of hyoid - Stylohyoid ligament	- Muscles of facial expression - Occipitofrontalis muscle - Stapedius - Stylohyoid - Posterior belly of digastric - Platysma	Stapedial artery	Facial nerve (VII)
3 rd	- Greater cornu and lower part of the body of hyoid bone	- Stylopharyngeus muscle.	Common Carotid artery	Glossopharyngeal nerve (IX)
4 th	- Thyroid cartilage.	- Cricothyroid muscle.	Subclavian Artery & aortic arch	Superior laryngeal nerve (Vagus nerve) (X)
6 th	Laryngeal cartilages (Arytenoid, Cricoid. Corniculate & Cuneiform).	- All muscles of larynx except cricothyroid muscle. - Constrictor muscles of the pharynx. All muscles of the palate (except tensor palati).	Pulmonary Arteries & ductus arteriosus	Recurrent laryngeal nerve (Vagus nerve) (X)

Derivatives of pharyngeal arches



Derivatives of 4 pharyngeal pouches:

- They are grooves lined with endoderm.
- Separate internally pharyngeal arches from each other.
- **First pouch:** gives rise to Middle ear cavity, Auditory (Eustachian) tube & Mastoid antrum.
- **Second pouch:** gives palatine tonsils & tonsillar fossa.
- **Third pouch:** gives rise to Inferior parathyroid gland & Thymus gland.
- **Fourth pouch:** gives rise to Superior parathyroid gland & Ultimobranchial body gives parafollicular cells of thyroid gland.

N.B. the 5th pharyngeal pouch is absent or rudimentary.

Derivatives of 4 pharyngeal clefts:

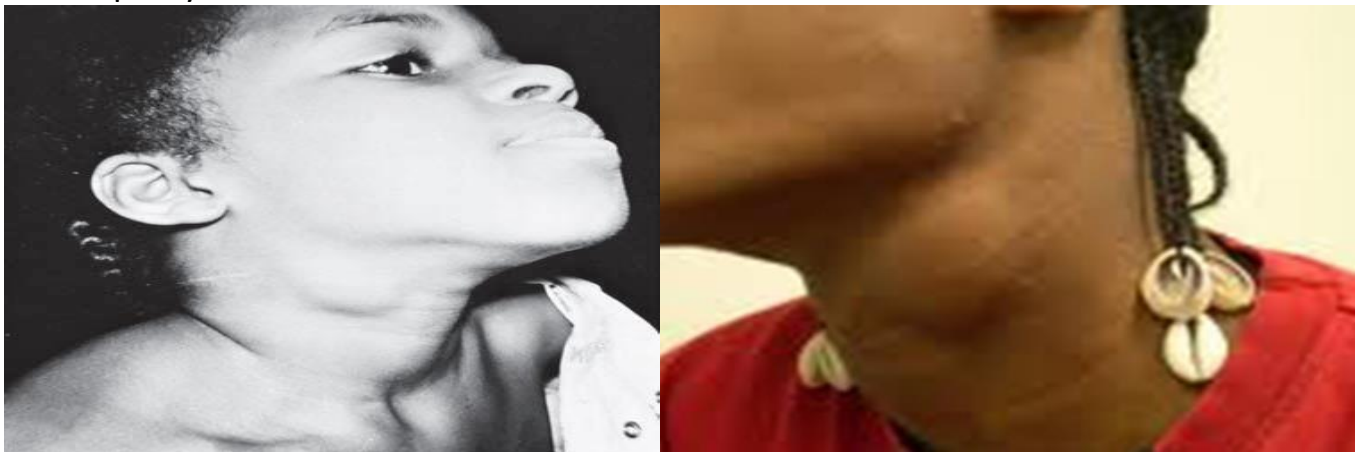
- They are grooves covered by surface ectoderm.
- Separate externally pharyngeal arches from each other.
- **First cleft:** gives rise to the **external auditory meatus**.
- 2nd, 3rd, 4th clefts open in a single cavity called **cervical sinus** on the side of neck, which later disappears.

Derivatives of pharyngeal membranes:

- **First pharyngeal membrane** gives rise to the **tympanic membrane**. 2nd, 3rd, 4th pharyngeal membranes disappear.

Congenital anomalies of pharyngeal arches:

- **Branchial cyst:** is due to persistence of cervical sinus on the lateral side of neck in front of the sternomastoid muscle.
- **Branchial sinus:** when the cyst opens outside or into the pharynx.
- **Branchial fistula:** when the sinus opens externally and internally, connects the pharynx with the outside.



Branchial cyst